Permutation inference for displaced/non-displaced EEG decoding

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1 Question and fixed primary analysis

Does the fixed eight-channel, 100 ms shrinkage-LDA decoder exceed a participant-specific label-shuffled chance distribution under the existing whole-recording validation? The 23 non-pilot participants form the primary evaluation. mle01 and spe30 informed feature/model selection and remain explicitly exploratory. The classification research note reports the observed decoder performance.

The primary pipeline uses continuous 1–20 Hz bandpass filtering without FORCe. It reuses the accepted trial IDs, event alignment, baseline and quality checks from DEC_02, described in the classification research note. Seven 100 ms means per channel over $[0.2, 0.9)$ s provide 56 features. Standardisation and shrinkage LDA (equal class priors, automatic covariance shrinkage) are fitted on training recordings only. No preprocessing or model tuning is repeated or changed here.

2 Permutation method and statistic

We draw 1,999 random label permutations independently within each recording, preserving its D/ND counts. Each permutation refits the models using exactly the existing leave-one-recording-out folds, separately within participant/session. Test predictions are pooled within a session before computing balanced accuracy (mean D/ND recall). Participant statistics average eligible session scores equally; each participant's null statistic uses the same aggregation. Exclusions remain rle13/pre and vco27/pre, whose training folds cannot support the protocol. The analysis covers 73 eligible sessions and 30,210 evaluated trials.

The one-sided Monte Carlo value is

$$p = \frac{1 + \#\{b : T_b \geq T_{\text{obs}}\}}{1 + 1999},$$

with ties included and minimum attainable $p = 0.0005$. Chance_{95} is the empirical 95th percentile of the null (upper order statistic), an unadjusted descriptive threshold. Holm adjustment controls family-wise error across the 23 primary participant tests. The DEC web report gives unadjusted session-level values for exploratory checks.

The DEC web report also includes central LDA, the all-eight-channel binned tree, and the Graz-style top-100 amplitude tree. The Graz-style tree ranks 8,000 candidate features (eight channels $\times$ 1,000 time samples per epoch) and selects the top 100 using training data within each permutation and fold. The three secondary models give $23 \times 3 = 69$ participant-level tests in a separate Holm family. Each participant–model test combines eligible sessions as above. The two exploratory pilots each contribute four model tests, giving a separate eight-test family. We verify the observed held-out predictions for all four models against DEC_02 and DEC_03.

3 Primary participant results

Eight-channel shrinkage LDA exceeded the within-recording label-shuffle null after Holm correction in 23/23 non-pilot participants. The unadjusted participant-specific 95th-percentile chance thresholds ranged from 52.6% to 54.9%. The Holm-corrected p-values determine significance after adjustment. Raw primary p-values ranged from 0.0005 to 0.0065. The Monte Carlo minimum of 0.0005 occurred for 22/23 participants. All primary Holm-adjusted p-values were 0.0115.

Participant	Sessions	Trials	BA (%)	Chance_{95} (%)	p	p_{Holm}
ara13	3	1,325	84.2	53.1	0.0005	0.0115
bal08	3	1,261	87.4	52.9	0.0005	0.0115
bpe06	3	1,140	70.3	53.2	0.0005	0.0115
bro19	3	816	79.2	54.8	0.0005	0.0115
cna07	3	994	84.8	53.5	0.0005	0.0115
cve01	3	1,405	86.6	52.8	0.0005	0.0115
den12	3	1,364	71.1	52.7	0.0005	0.0115
eab31	3	1,423	66.3	52.9	0.0005	0.0115
emo24	3	884	74.4	53.9	0.0005	0.0115
fbo09	3	1,332	82.3	53.0	0.0005	0.0115
gba12	3	1,414	82.6	53.0	0.0005	0.0115
gcr27	3	1,439	73.4	53.3	0.0005	0.0115
lbo12	3	1,265	71.0	53.1	0.0005	0.0115
ldu23	3	1,275	72.6	53.3	0.0005	0.0115
meb17	3	1,010	72.1	53.4	0.0005	0.0115
mka24	3	1,398	77.2	52.9	0.0005	0.0115
nfu22	3	1,435	80.8	52.6	0.0005	0.0115
nsh23	3	1,442	71.8	52.7	0.0005	0.0115
rle13	2	887	86.1	53.6	0.0005	0.0115
sga12	3	1,121	72.5	53.2	0.0005	0.0115
sto20	3	1,291	71.9	53.2	0.0005	0.0115
vco27	2	612	56.8	54.4	0.0065	0.0115
yle21	3	1,112	79.6	54.9	0.0005	0.0115

Table 1: Non-pilot participants. BA is the equal-session mean balanced accuracy. Chance_{95} uses the identical statistic under within-recording label permutations. Holm values refer to the 23-test primary family.

Participant	Sessions	Trials	BA (%)	Chance_{95} (%)	p	p_{Holm}
mle01	3	1,179	75.9	53.0	0.0005	0.0040
spe30	3	1,386	85.8	52.7	0.0005	0.0040

Table 2: Exploratory pilots, shown separately. Their Holm values belong to the eight-test pilot/model family; they are not independent validation of pilot-driven choices.

The DEC web report contains the secondary model tables and unadjusted session checks.

4 Observed performance and chance thresholds

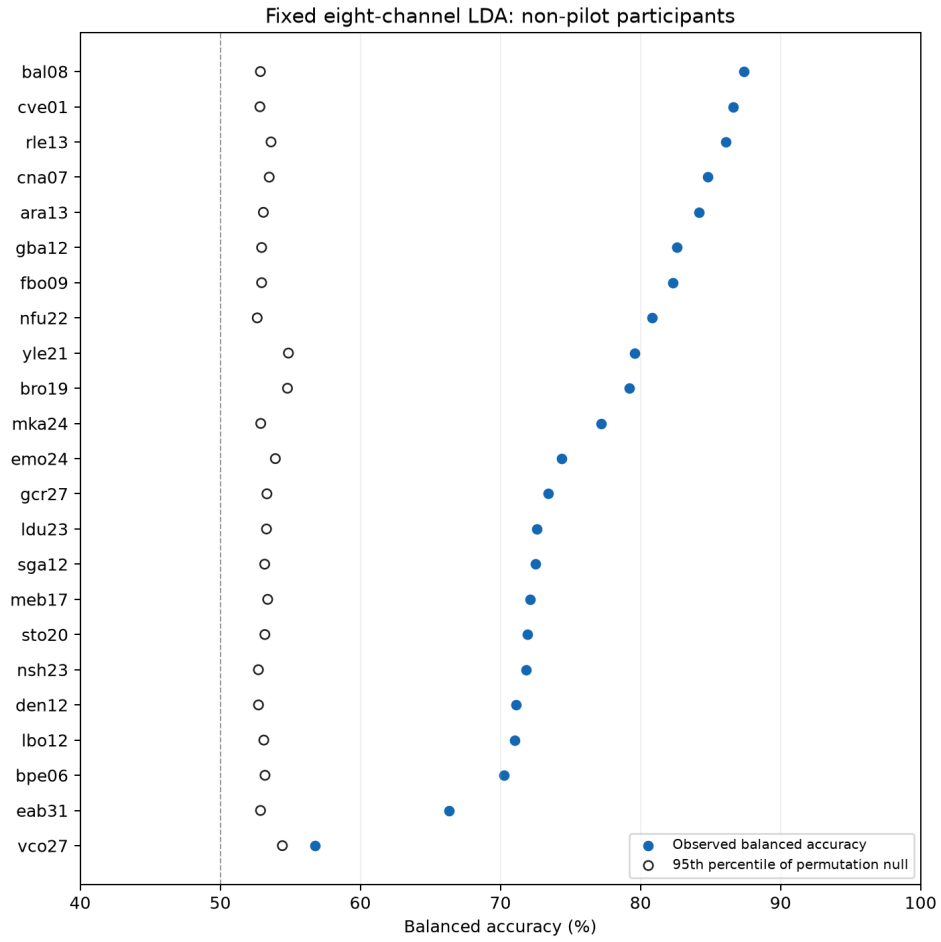


Figure 1: Observed participant scores and unadjusted permutation-derived thresholds for the primary LDA. The threshold markers are not confidence intervals or family-wise corrected decision thresholds.

5 Interpretation and limits

The null assumes D/ND labels are exchangeable within each recording. Blocking by recording preserves recording-level class proportions but does not establish exchangeability in the presence of unmodelled temporal structure. This is inference conditional on the existing trial inclusion and fixed personalised decoder, not a test of generalisation of a trained model to new participants.

Above-null discrimination supports useful label-related information. It does not identify an ErrP, establish error awareness, or rule out visual, movement-related or residual artifact contributions. Waveform morphology, timing and scalp distribution remain complementary evidence.

Method references. Ojala and Garriga, *Permutation tests for studying classifier performance*, JMLR 11 (2010), 1833–1863. Phipson and Smyth, *Permutation p-values should never be zero*, SAGMB 9 (2010), Article 39, doi:10.2202/1544-6115.1585. The implementation follows the grouped-shuffle and +1 Monte Carlo convention documented by scikit-learn.